

SQL 2

1. Display the count of the number of rows in a table.

The screenshot shows a SQL query window titled "SQLQuery1.s... \praty (85))*" with the following SQL code:

```
1  -- Display the count of the number of rows in a table.  
2  
3  Select Count(*) as PCnt from Provider  
4
```

Below the query window, the "Results" tab is active, displaying a single row with the column "PCnt" and the value "27". The status bar indicates "100 %", "No issues found".

PCnt
27

2. Delete a specified record in a table and prove that it is deleted.

The screenshot shows a SQL query window titled "SQLQuery2.s... \praty (52))*" with the following SQL code:

```
1  -- Delete a specified record in a table and prove that it is deleted.  
2  
3  Select Count(*) as PTIDCnt from ProviderType -- we selected from ProviderType, and made a column named PTIDCnt  
4  go  
5  Delete from ProviderType -- we deleted ProviderType whoes ProviderTypeID was 5 or 6  
6  Where ProviderTypeID = 5 or ProviderTypeID = 6  
7  go  
8  Select * from ProviderType --we proved that ProviderTypeID 5 or 6 were indeed deleted  
9  Where ProviderTypeID = 5 or ProviderTypeID = 6
```

Below the query window, the "Results" tab is active, displaying a single row with the column "PTIDCnt" and the value "6". The status bar indicates "100 %", "No issues found".

PTIDCnt
6

ProviderTypeID	ProviderType	ProTypeDesc
----------------	--------------	-------------

3. Two table join that creates a view.

The screenshot shows a SQL query window with the following text:

```
-- Two table join that creates a view.  
Create view vw_GetProviderAndType as -- creates a view on view subdirectory  
Select FName, LName, ProviderType  
From Provider P, ProviderType PT  
Where P.ProviderTypeID = PT.ProviderTypeID  
  
Select FName from vw_GetProviderAndType -- gets teh content from vw_GetProviderTypeID
```

The results pane shows a list of 25 rows, each with an FName:

FName
1 James
2 Emily
3 Michael
4 Sarah
5 David
6 Laura
7 Robert
8 Jessica
9 Daniel
10 Amanda
11 Matthew
12 Olivia
13 Andrew
14 Stephanie
15 Christopher
16 Nicole
17 Brian
18 Rachel
19 Kevin
20 Megan
21 Jonathan
22 Hannah
23 Eric
24 Victoria
25 Steven

4. Three table join qualified with a key value from one view and one other table.

The screenshot shows a SQL query window with the following text:

```
-- Three table join qualified with a key value from one view and one other table.  
  
Select ProviderType, LName, Ailment  
From vw_GetProviderAndType V, PatientAilment PA, Ailment A  
Where V.ProviderID = PA.ProviderID  
and A.AilmentID = PA.AilmentID  
and LName like '%p%' -- will list all the letters starting and ending with a 'p'
```

The results pane shows a table with 9 rows and 3 columns: ProviderType, LName, and Ailment.

ProviderType	LName	Ailment
1 Physician	Patel	Seasonal Allergies
2 Physician	Lopez	Arthritis
3 Physical Therapist	Priest	Vertigo
4 Registered Nurse	Peterson	Sleep Apnea
5 Phlebotomist	Powell	Gastroesophageal Reflux Disease
6 Physician	Patel	Depression
7 Physician	Lopez	Asthma
8 Physical Therapist	Priest	Eczema
9 Registered Nurse	Peterson	Pneumonia

5. Use five SQL functions (such as SUM). This should be five queries with at least one function for each query, and not the same functions as you used in previous assignments.

SQLQuery8.s... \praty (89))* SQLQuery7.s...ot connected* SQLQuery6.s...ot connected*

```
1  /*
2  Use five SQL functions (such as SUM).
3  This should be five queries with at least one function for each query,
4  and not the same functions as you used in previous assignments.
5  */
6
7  Select ProviderType, Len(ProviderType) as Typelength
8  From ProviderType
9
```

100 % No issues found

Results Messages

	ProviderType	Typelength
1	Physician	9
2	Registered Nurse	16
3	Physical Therapist	18
4	Phlebotomist	12
5	XRay Tech	9
6	CardioVascular Surgeon	21

SQLQuery8.s... \praty (89))* SQLQuery7.s...ot connected* SQLQuery6.s...ot connected*

```
3  This should be five queries with at least one function for each query,
4  and not the same functions as you used in previous assignments.
5  */
6
7  SELECT ProviderTypeID, COUNT(*) AS UsageCount
8  FROM Provider
9  GROUP BY ProviderTypeID
10 ORDER BY UsageCount DESC
11
```

100 % No issues found

Results Messages

	ProviderTypeID	UsageCount
1	1	9
2	2	6
3	3	6
4	4	4
5	NULL	2

SQLQuery8.s...\praty (89))* SQLQuery7.s...ot connected* SQLQuery6.s...ot connected*

1

2

3

4

5

6

7

8

9

10

/*

Use five SQL functions (such as SUM).

This should be five queries with at least one function for each query,

and not the same functions as you used in previous assignments.

*/

SELECT UPPER(FName) as FirstNameUpper

From Provider

100 % No issues found

Results Messages

	FirstNameUpper
1	JAMES
2	EMILY
3	MICHAEL
4	SARAH
5	DAVID
6	LAURA
7	ROBERT

SQLQuery8.s...\praty (89))* SQLQuery7.s...ot connected* SQLQuery6.s...ot connected*

1

2

3

4

5

6

7

8

9

10

/*

Use five SQL functions (such as SUM).

This should be five queries with at least one function for each query,

and not the same functions as you used in previous assignments.

*/

SELECT MAX(ProviderTypeID) as MaxProviderTypeID

From ProviderType

100 % No issues found

Results Messages

	MaxProviderTypeID
1	6

SQLQuery8.s... \praty (89))* SQLQuery7.s...ot connected* SQLQuery6.s...ot connected*

```

1  /*
2  Use five SQL functions (such as SUM).
3  This should be five queries with at least one function for each query,
4  and not the same functions as you used in previous assignments.
5  */
6
7  SELECT Min(ProviderTypeID) as MinProviderTypeID
8  From ProviderType
9

```

100 % No issues found

Results Messages

	MinProviderTypeID
1	1

6. Create a SQLstored procedure to insert values into a row using variables passed in on execution.

loq\SQLXPRES...ProviderType SQLQuery6.s... \praty (53))* SQLQuery2.sq... \praty (52))* SQLQuery4.s...ot connected*

```

1  -- Create a SQLstored procedure to insert values into a row using variables passed in on execution.
2
3  Create Proc spu_InsertProviderTypeDynamic
4  @ProviderType nvarchar(50), @ProTypeDec nvarchar(Max)
5  as
6
7  Insert into ProviderType
8  (ProviderType, ProTypeDesc)
9  Values
10  (@ProviderType, @ProTypeDec)
11
12  Exec spu_InsertProviderTypeDynamic
13  'CardoVascular Surgeon',
14  'Heart Doctor'

```

100 % No issues found

Messages

(1 row affected)

Completion time: 2026-03-04T18:23:38.7977652-05:00

7. Write the SQL to create a table, key, and index. This must not be generated SQL.
Example after #12.

```
SQLQuery8.s...praty (89))*  SQLQuery7.s...ot connected*  SQLQuery6.s...ot connected*
1  -- Write the SQL to create a table, key, and index. This must not be generated SQL. Example after #12.
2  -- Create Student table to store basic student information
3
4  CREATE TABLE Student
5  (
6  StudentID int NOT NULL,
7  FName nvarchar(20) NOT NULL,
8  LName nvarchar(30) NOT NULL,
9  MName nvarchar(20) NULL,
10 -- Defining a primary key constraint to ensure that each StudentID is unique
11 CONSTRAINT PK_Student PRIMARY KEY CLUSTERED
12
13 (StudentID ASC)-- The primary key is clustered and sorted in ascending order
14 ) -- key
15 ON [PRIMARY] -- Specifying the filegroup for the index
16
```

8. Create a query that uses Group By

```
SQLQuery7.s...praty (52))*  loq\SQLXPRES...ProviderType  SQLQuery6.sq..
1  -- Create a query that uses Group By
2
3  Select LName, FName, Count(PatientID) as PatientCnt
4  From Provider P, PatientAilment PA
5  Where P.ProviderID = PA.ProviderID
6  Group By LName, FName
7  Order By PatientCnt Desc
8
```

100 % No issues found

Results Messages

	LName	FName	PatientCnt
1	Ramirez	Hannah	2
2	Wilson	James	2
3	Morris	Jessica	2
4	Long	Jonathan	2
5	Flores	Kevin	2
6	Hernandez	Laura	2
7	Sanders	Matthew	2
8	Russell	Megan	2
9	Reed	Michael	2
10	Ward	Nicole	2
11	Price	Olivia	2
12	Peterson	Rachel	2

9. Demonstrate the use of distinct, ascending, descending in a query. The result set must have at least 7 rows.

The first screenshot shows a SQL query in SQL Server Enterprise Manager. The query is as follows:

```
-- Demonstrate the use of distinct, ascending, descending in a query. The result set must have at least 7 rows.

Select FName -- Selecting only First Name
From Provider -- Specifying from where to select
Order By FName Desc -- Instruction given as to arrange them in Descending order
```

The results pane shows 10 rows of first names in descending order:

FName
Victoria
Steven
Stephanie
Sarah
Samantha
Robert
Rachel
Olivia
Nicole
Michael

The second screenshot shows the same query but with 'Order By FName Asc' instead of 'Desc'. The results pane shows 11 rows of first names in ascending order:

FName
Amanda
Andrew
Bill
Bitan
Christopher
Daniel
David
Emily
Eric
Hannah
James

10. Create a query that demonstrates an inner join (includes join in statement)

The screenshot shows a SQL query in SQL Server Enterprise Manager. The query is as follows:

```
-- Create a query that demonstrates an inner join (includes join in statement)

Select LName, ProviderType
From Provider P
Inner Join ProviderType PT
On P.ProviderTypeID = PT.ProviderTypeID
```

The results pane shows 9 rows of data:

LName	ProviderType
Wilson	Physician
Carter	Registered Nurse
Reed	Physician
Nguyen	Physical Therapist
Patel	Physician
Hernandez	Registered Nurse
Kim	Phlebotomist
Morris	Physical Therapist
Lopez	Physician

11. Create a query that demonstrates a left outer join (includes join in statement)

SQLQuery8.s...praty (89)* SQLQuery7.s...ot connected* SQLQuery6.s...ot connected*

```
1 -- Create a query that demonstrates a left outer join (includes join in statement)
2
3 Select LName, ProviderType
4 From Provider P
5 Left Join ProviderType PT
6 On P.ProviderTypeID = PT.ProviderTypeID
```

150 % No issues found

Results Messages

	LName	ProviderType
1	Wilson	Physician
2	Carter	Registered Nurse
3	Reed	Physician
4	Nguyen	Physical Therapist
5	Patel	Physician
6	Hernandez	Registered Nurse
7	Kim	Phlebotomist
8	Morris	Physical Therapist
9	Lopez	Physician
10	Brooks	Registered Nurse

12. Create a query that demonstrates a right outer join (includes join in statement)

SQLQuery8.s...praty (89)* SQLQuery7.s...ot connected* SQLQuery6.s...ot connected*

```
1 -- Create a query that demonstrates a right outer join (includes join in statement)
2
3 Select LName, ProviderType
4 From Provider P
5 Right Join ProviderType PT
6 On P.ProviderTypeID = PT.ProviderTypeID
```

150 % No issues found

Results Messages

	LName	ProviderType
1	Wilson	Physician
2	Reed	Physician
3	Patel	Physician
4	Lopez	Physician
5	Sanders	Physician
6	Howard	Physician
7	Cox	Physician
8	Long	Physician
9	Coleman	Physician
10	Carter	Registered Nurse